The Journal of Open Source Software

SERGHEI-QGIS: A QGIS Plugin Suite for End-to-End Flood Simulation Workflows with the SERGHEI-SWE Model

Shahin Khosh Bin Ghomash [1] and **Daniel Caviedes-Voullième** [1,2]

1 Chair for Environmental Fluid Dynamics and Modeling, TUD Dresden University of Technology, Dresden, Germany **2** Simulation and Data Lab Terrestrial Systems, Institute of Bio- and Geosciences Agrosphere and Jülich Supercomputing Centre, Forschungszentrum Jülich, Jülich, Germany

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 04 August 2026
Published: unpublished

Summary

Physically-based two-dimensional flood modelling with the shallow-water equations (SWE) requires a chain of technically demanding steps: preparing geospatial inputs, configuring the model, managing high-performance computing (HPC) resources, and post-processing results. The SERGHEI-SWE model (Caviedes-Voullième et al., 2023) provides a high-performance, GPU-accelerated solver for these problems, but no integrated graphical tooling existed to support the full workflow from within a geographic information system (GIS) environment.

`SERGHEI-QGIS` is a suite of five QGIS plugins and a standalone Python library that covers every stage of a SERGHEI-SWE simulation directly inside QGIS. The suite allows users to prepare model inputs from standard geospatial datasets, acquire, generate, and process rainfall forcing, compile and submit simulation jobs to remote HPC clusters, reload existing cases, and visualise, animate, and export results, all without leaving the QGIS desktop environment. The plugins are compatible with QGIS 3.34–3.40 (PyQt5) and QGIS 4.0+ (PyQt6). The source code is openly available on GitLab (Khosh Bin Ghomash, 2026).

Statement of need

Deploying SERGHEI-SWE requires proficiency across GIS data handling and processing, Linux based HPC environments, CMake-based build systems, SLURM job scheduling, NetCDF data management and visualization. This breadth of prerequisites creates an entry barrier for researchers and practitioners whose expertise lies in hydrology or hydraulics rather than scientific computing workflows or research software engineering. Contrary to other fields in Earth system modelling, hydrology and hydraulics has traditionally relied more on GUIs and workstation solvers rather than scientific workflows on HPC systems.

Existing QGIS plugin ecosystems provide powerful geospatial tools but do not address the specific input format requirements of SERGHEI-SWE, nor do they handle the issues of model compilation and HPC job management. Standalone scripts and command-line workflows often require manual coordination between tools that were not designed to interoperate. Even fully automated CLI operated workflows may arguably still be intransparent and unfriendly for users not used to HPC environments and complex CLI processes.

`SERGHEI-QGIS` addresses this gap with a coherent, GUI-driven interface that integrates all simulation stages. This reduces the technical overhead for new users, promotes reproducibility by encoding best-practice workflows, and makes the SERGHEI modelling chain accessible to a broader community of flood researchers and practitioners.

State of the field

Several QGIS plugins support flood-related workflows, including RiverGIS (RiverGIS Team, 2018) for HEC-RAS preprocessing and the TUFLOW Plugin (Ryan et al., 2026). These target their respective model formats and provide no pathway to GPU-accelerated, performance-portable solvers (Caviedes-Voullième et al., 2023). On the other hand, QGIS Processing Toolboxes (e.g., SAGA, GRASS) offer generic terrain analysis but not model-specific input preparation or HPC integration.

On the HPC side, tools such as Cylc (Oliver et al., 2019) provide workflow orchestration for climate simulations but require separate infrastructure and are oriented towards automated pipelines rather than interactive, case-specific scientific use.

`SERGHEI-QGIS` occupies a distinct niche by bringing model-specific preprocessing, operational rainfall acquisition, HPC job management, and simulation result visualisation together in a single QGIS-native environment, rather than leaving them to separate tools.

Software design

`SERGHEI-QGIS` comprises a top-level toolbox plugin (`SERGHEI_tools`) and five functional plugins, backed by a standalone, QGIS-independent Python library (`serghei_model_setup.py`) that encapsulates all SERGHEI input construction:

- **Model Setup** builds the complete set of SERGHEI-SWE inputs: digitial terrain model, maps of hydraulic roughness, soil and land use properties, initial and boundary conditions, and observation points and lines. These are built from standard QGIS layers, writing either ESRI-ASCII Grid files or a single CF-compliant NetCDF.
- **Rainfall Processing** acquires observed precipitation from a set of providers (DWD RADOLAN and RADKLIM, ICON-D2, MRMS, and CHIRPS), maps tabulated station series onto the model grid through rainfall-zone polygons, and generates synthetic space-time rainfall through several stochastic methods (Papalexiou, 2018; Paschalis et al., 2013) with selectable heavy-tailed marginals (Naveau et al., 2016), including a seasonally non-stationary, streaming multi-year generator (Peleg et al., 2017).
- **Compile/Run** manages the remote HPC processes over SSH—configuration, compilation, SLURM job submission, live monitoring, batch ensembles, and result retrieval—with presets for a set of known systems (Capella at ZIH/TUD (NHR Center at TU Dresden, 2024), JUWELS Booster (Alvarez, 2021) and JUPITER Booster (Jülich Supercomputing Centre, 2025) at JSC) and an embedded interactive terminal. Beyond these, an auto-detection mode probes an unknown cluster for a working compiler, MPI and CUDA toolchain and builds missing dependencies from source, and further modes target remote workstations without a scheduler and the user's own machine.
- **Result Visualiser** animates temporal output and loads static results as styled QGIS layers, and exports them for Google Earth (KMZ) or as annotated GIF and MP4.
- **Load Existing Model** reconstructs a case from its SERGHEI input files as styled QGIS layers, which can be handed back to Model Setup for modification and rebuilding.

All plugins share a two-class controller pattern, keep background work off the GUI thread through `pyqtSignal` connections, and run on both PyQt5 (QGIS 3.34–3.40) and PyQt6 (QGIS 4.0+) via a compatibility shim. Continuous integration guards both lines: the standalone library is unit-tested in isolation from the graphical layer, while every plugin module and interface definition is imported and instantiated inside the official QGIS container images for each supported version, with the QGIS nightly build run as an early warning. On the solver side no model version is bundled: SERGHEI is built from its public repository with the branch selectable, so the suite follows upstream development. The toolbox additionally provides one-click dependency installation, self-updating, and integrated, scoped error reporting. A manual documenting each tool and tab ships with the plugin and opens from the toolbox.

Research impact

`SERGHEI-QGIS` is developed at TUD Dresden University of Technology in direct support of ongoing research in computational hydrology, hydrodynamics and flood simulation. By integrating input preparation, rainfall acquisition, HPC job management, and result visualisation into a single QGIS-native workflow, the suite removes the technical overhead that often hampers the uptake of physically-based, GPU-accelerated flood modelling by users typically lacking substantial scientific computing or software engineering expertise. Nevertheless, since the underlying python scripts can also be executed in a programatic fashion, experienced users can easily integrate the implementations into broader computational workflows.

More broadly, it underpins research involving large-scale, GPU-accelerated inundation modelling on European HPC infrastructure, where the ability to manage compilation, job submission, and result retrieval from within the GIS environment substantially shortens the iteration cycle between scenario design and analysis. The suite has already been used to prepare, run, and post-process the simulations underpinning a study of high-resolution rain-on-grid hydrodynamic modelling of the 2021 Ahr catchment flood (Khosh Bin Ghomash & Caviedes-Voullième, 2026), and it currently supports several ongoing research projects at TU Dresden.

Beyond its current research applications, the suite lowers the barrier to entry for the SERGHEI-SWE model itself, making a state-of-the-art performance-portable solver accessible to hydrologists, civil engineers, and practitioners whose work centres on flood risk rather than computational tooling. By encoding established workflows and correct data-handling conventions directly into the interface, it also promotes and fosters reproducibility and reduces the scope for user error in operational flood-modelling studies.

Finally, by leveraging an entirely open source software stack, it also becomes a prime tool for teaching and training, facilitating access for undergraduate and graduate students to interact with SERGHEI with significantly lower technical learning curves.

AI usage disclosure

Generative AI (Anthropic's Claude Opus, accessed through Claude Code) was used for assistance during the development of `SERGHEI-QGIS`. Its use spanned assistance in coding within the plugin modules and some plugin documentation drafting. All architectural and scientific decisions (the design of the plugin suite, the SERGHEI input conventions it encodes, and the interpretation of the solver's behaviour) and the majority of the coding were done by the authors. The authors reviewed, edited and validated every AI-assisted output, and are responsible for the correctness of the software and of this paper.

Acknowledgements

The authors gratefully acknowledge the Earth System Modelling (ESM) project of the Helmholtz Association for computing time on the ESM partition of the JUWELS supercomputer (Alvarez, 2021) at the Jülich Supercomputing Centre (JSC), through the compute time project Environmental Hydrodynamics, runoff and transport across scales with the SERGHEI model (EHRTAS) and the ESM Test project.

They further thank the Gauss Centre for Supercomputing e.V. (GCS) for computing time on the supercomputer JUPITER at JSC via the John von Neumann Institute for Computing, and the JUPITER Research and Early Access Program (JUREAP) for early access to JEDI and JUPITER, which enabled the preparatory work for deploying SERGHEI. JUPITER was supported by the EuroHPC JU and GCS through funding by the European Commission, the German Federal Ministry of Research, Technology and Space, and the Ministry of Culture and Science of the State of North Rhine-Westphalia.

The authors further gratefully acknowledge the computing time made available to them on the high-performance computer Capella (NHR Center at TU Dresden, 2024) at the NHR Center of TUD Dresden University of Technology. This center is jointly supported by the Federal Ministry of Research, Technology and Space of Germany and the state governments participating in the NHR (https://www.nhr-verein.de/unsere-partner).

References

Alvarez, D. (2021). JUWELS cluster and booster: Exascale pathfinder with modular supercomputing architecture at juelich supercomputing centre. *Journal of Large-Scale Research Facilities*, *7*, A183. https://doi.org/10.17815/jlsrf-7-183

Caviedes-Voullième, D., Morales-Hernández, M., Norman, M. R., & Özgen-Xian, I. (2023). SERGHEI (SERGHEI-SWE) v1.0: A performance-portable high-performance parallel-computing shallow-water solver for hydrology and environmental hydraulics. *Geoscientific Model Development*, *16*, 977–1008. https://doi.org/10.5194/gmd-16-977-2023

Jülich Supercomputing Centre. (2025). *JUPITER technical overview*. Forschungszentrum Jülich, for the EuroHPC Joint Undertaking. https://www.fz-juelich.de/en/jsc/jupiter/tech

Khosh Bin Ghomash, S. (2026). *SERGHEI-QGIS: A QGIS plugin suite for SERGHEI-SWE flood simulations*. https://gitlab.com/serghei-model/serghei-qgis-plugin

Khosh Bin Ghomash, S., & Caviedes-Voullième, D. (2026). High-resolution rain-on-grid hydrodynamic modelling can replace hydrological models for catchment-scale flood simulation: The case of the 2021 Ahr catchment flood. *EGUsphere (Preprint), Natural Hazards and Earth System Sciences*. https://doi.org/10.5194/egusphere-2026-3652

Naveau, P., Huser, R., Ribereau, P., & Hannart, A. (2016). Modeling jointly low, moderate, and heavy rainfall intensities without a threshold selection. *Water Resources Research*, *52*(4), 2753–2769. https://doi.org/10.1002/2015WR018552

NHR Center at TU Dresden. (2024). *GPU cluster Capella*. ZIH HPC Compendium, Center for Information Services and High Performance Computing (ZIH), TUD Dresden University of Technology. https://compendium.hpc.tu-dresden.de/jobs_and_resources/capella/

Oliver, H., Shin, M., Matthews, D., Sanders, O., Bartholomew, S., Clark, A., Fitzpatrick, B., Haren, R. van, Hut, R., & Drost, N. (2019). Workflow automation for cycling systems: The Cylc workflow engine. *Computing in Science & Engineering*, *21*(4), 7–21. https://doi.org/10.1109/MCSE.2019.2906593

Papalexiou, S. M. (2018). Unified theory for stochastic modelling of hydroclimatic processes: Preserving marginal distributions, correlation structures, and intermittency. *Advances in Water Resources*, *115*, 234–252. https://doi.org/10.1016/j.advwatres.2018.02.013

Paschalis, A., Molnar, P., Fatichi, S., & Burlando, P. (2013). A stochastic model for high-resolution space-time precipitation simulation. *Water Resources Research*, *49*(12), 8400–8417. https://doi.org/10.1002/2013WR014437

Peleg, N., Fatichi, S., Paschalis, A., Molnar, P., & Burlando, P. (2017). An advanced stochastic weather generator for simulating 2-D high-resolution climate variables. *Journal of Advances in Modeling Earth Systems*, *9*(3), 1595–1627. https://doi.org/10.1002/2016MS000854

RiverGIS Team. (2018). *RiverGIS: A QGIS plugin for creating HEC-RAS flow model geometry from spatial data* (Version 3.0). https://plugins.qgis.org/plugins/rivergis/

Ryan, P., Symons, E., & Monhartova, P. (2026). *TUFLOW QGIS plugin: Utilities for TUFLOW flood and coastal simulation in QGIS* (Version 2026.0.18). https://plugins.qgis.org/plugins/tuflow/